MEDICAL INSURANCE COST PREDICTION

Problem Statement

Medical insurance costs vary significantly based on individual factors such as age, body mass index (BMI), smoking habits, family size, and geographic location. Predicting insurance charges accurately is essential for insurance companies, healthcare providers, and customers for cost planning and risk assessment.

This project focuses on developing a machine learning-based predictive model to estimate individual medical insurance charges based on beneficiary demographic and health-related attributes. By analyzing personal and lifestyle factors, the project aims to automate insurance cost prediction and support data-driven healthcare financial planning.

Objectives:

- Predict individual medical insurance charges
- Analyze the impact of demographic and lifestyle factors on insurance costs
- Build an effective regression model
- Evaluate predictive performance using R-squared metrics
- Support healthcare cost estimation and insurance planning

Dataset Used

Dataset Context:

The dataset is based on data used in "Machine Learning with R" by Brett Lantz. The dataset was cleaned and reformatted for machine learning applications.

Dataset Features:

Input Variables:

1. Age – Age of the primary beneficiary
2. Sex – Gender (female, male)
3. BMI – Body Mass Index
4. Children – Number of dependents covered by insurance
5. Smoker – Smoking status
6. Region – Residential region in the US:
    - Northeast
    - Southeast

   - Southwest
   - Northwest

Output Variable:

7. Charges (Individual medical costs billed by health insurance)

Approach / Methodology

Data Preparation:

- Loaded medical insurance dataset
- Selected demographic and health-related features as input variables
- Used charges as the target variable
- Performed data cleaning and preprocessing
- Encoded categorical variables
- Split data into training and testing datasets

Machine Learning Model Used:

Linear Regression:

Linear Regression is a supervised machine learning algorithm used to predict continuous numerical outcomes by modeling the relationship between dependent and independent variables.

Reason for Selection:

- Suitable for continuous value prediction
- Easy to interpret
- Effective baseline regression model
- Provides clear statistical evaluation through R-squared scores

Evaluation Metric

R-Squared Score (R^2):

R^2 measures how well the independent variables explain the variance in the target variable.

- $R^2 = 1 \rightarrow$ Perfect prediction
- $R^2 = 0 \rightarrow$ No predictive capability

Results and Evaluation

Linear Regression Performance

Training Data R-Squared Score:

0.7515

Testing Data R-Squared Score:

0.7447

Performance Analysis:

- The model explains approximately 75.15% of variance in training data
- The model explains approximately 74.47% of variance in testing data
- Similar train and test scores indicate:
    - Good generalization
    - Minimal overfitting
    - Stable model performance

Interpretation:

- The model performs reasonably well for medical cost prediction
- Important factors such as smoking, BMI, and age likely contribute significantly
- Linear Regression provides reliable baseline predictive capability

Conclusion

This project successfully developed a machine learning-based medical insurance cost prediction system using beneficiary demographic and lifestyle data.

Linear Regression provided:

- Strong baseline regression performance
- Good generalization
- Stable prediction capability
- Easy interpretability

Thus, Linear Regression is an effective model for estimating medical insurance costs based on available personal and health-related factors.